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Simulation-based evaluation of occupancy on energy consumption of multi-scale residential building archetypes

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ABSTRACT

Building archetypes are commonly used to identify areas with energy-inefficient buildings and estimate energy demand at different spatial scales. However, these archetypes often rely on conventional occupancy schedules, which are fixed and uniform throughout the year. This study proposes a new methodology that uses data from the Time Use Survey (TUS) to extract realistic occupancy profiles. The occupancy profiles are used to model the heat gains due to occupancy along with their interaction with lighting and heating systems. The developed occupancy schedules take into account factors such as the number of occupants, dwelling types, the month of the year, and the day of the year. Energy simulations were conducted to evaluate the impact of realistic occupancy schedules on multi-scale residential building archetypes at different spatial scales (postcode, county, and national). The results showed that the developed occupancy schedules were significantly different from conventional schedules, resulting in approximately 8%–10% variations in annual energy demand at various spatial scales compared to the base case. Monthly heating and electricity consumption also showed variations of approximately 25% and 32%, respectively, compared to the base case. With more granularity, the daily variations in heating and electricity consumption were reported as 50% and 12%, respectively, compared to the base case with standard and conventional schedules. These results highlight the importance of incorporating more realistic occupancy schedules in building archetypes for more reliable energy predictions at different spatial and temporal scales. Occupancy-integrated archetypes can help policymakers, local authorities, and urban planners to make informed decisions based on actual data and develop effective energy recommendations for direct retrofit investment and policies.

1. Introduction

According to a report from the United Nations, the global urban population, consisting of 3.42 billion individuals, has surpassed the rural population, which is 3.41 billion [1]. The urban building sector is the largest energy consumer worldwide and contributes significantly to greenhouse gas (GHG) emissions [2]. As urbanisation continues to increase, the environment faces more stress, which highlights the need for sustainability to be integrated into effective urban planning within the construction sector. It is projected

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Acronyms

ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BEPS	Building Energy Performance Simulation
BER	Building Energy Rating
DEAP	Dwelling Energy Assessment Procedure
EPC	Energy Performance Certificates
HETUS	Harmonised European Time Use Survey
HVAC	Heating, Ventilation and Air Conditioning
IEQ	Indoor Environmental Quality
SEAI	Sustainable Energy Authority of Ireland
TUS	Time Use Survey

that urbanisation levels in Europe will rise from 74% in 2019 to 83.7% by 2050 [3]. In response, the European Union has set a target to reduce GHG emissions by 55% compared to 1990 levels by 2030 [4]. Policymakers aim to achieve this goal by improving the energy efficiency of existing buildings. However, it is predicted that 75% of the current inefficient building stock will still exist in 2050 [5]. Therefore, retrofitting existing buildings is crucial for achieving low carbon emissions and aligning with the Paris Agreement targets [6].

An archetype is a virtual 3D building model that represents a group of buildings with similar characteristics [7]. Numerous data-driven machine learning techniques are available to develop building archetypes using various datasets, such as Energy Performance Certificates (EPC), to identify different archetypes and corresponding input parameters for classification and identification purposes [8–11]. However, when archetypes estimate energy demand at different spatial and temporal scales, the calculated results often do not match real-world data [12–14]. One primary reason for this is the use of standard and uniform occupancy schedules throughout the year [15]. Many studies develop archetypes based on dwelling type, age of construction, and heating system while neglecting occupancy and its stochastic behaviour to simplify the process [16–19]. A review by Brogger et al. [20] on building stock modelling speculates that capturing and understanding the habitual trends of occupancy behavioural changes is essential for building energy prediction. Clevenger et al. quantified the impact of occupancy behaviour on building performance and found that predicted energy consumption could vary by up to 150% if all occupancy-related input parameters are maximised or minimised [21]. This leads to unrealistic energy demand profiles at various temporal resolutions [22–24]. Neglecting realistic occupancy schedules has led to approximately 20% variation in energy savings due to the adoption of energy efficiency measures [25]. Fewer studies use qualitative occupancy-related information to quantify the impact of occupancy on building energy consumption. Various studies have concluded that energy savings of 10%–20% can be achieved by adjusting the occupancy presence/absence behaviour alone [26–28].

This study aims to develop realistic day-wise occupancy profiles by utilising the Time Use Survey (TUS) data to represent a large number of households. The novel methodology presented in this study extracts occupancy schedules based on various categorisation attributes, such as the type of dwelling, number of occupants, months of the year, and days of the week. The study provides a new set of profiles that represent larger households than standard and uniform schedules. These profiles are integrated with residential building archetypes to quantify the impact of occupancy on total energy consumption, including lighting, space heating, and domestic hot water, at various spatial and temporal scales. The study's results demonstrate that using standard and uniform occupancy schedules as input for dynamic building energy simulations can lead to inaccurate predictions. The methodology developed in this study is generic and can be applied to different locations. As a case study, we apply the methodology to the Irish residential building archetypes to estimate the impact of realistic occupancy behaviour on energy consumption of archetypes available at postcode, county and national scales.

The paper consists of the following sections: Section 2 provides a literature review. Section 3 describes the methodology used to develop occupancy profiles and occupancy-based archetypes. Section 4 introduces a case study to implement the devised methodology and includes the analysis of the results. Section 5 discusses the output of the approach. Finally, we end with conclusions and future work in Section 6.

2. Literature review

Cities contain thousands to millions of buildings and estimating the energy demand of each building for multi-scale analysis is a tedious and computationally heavy task [29]. However, energy estimation at a district, city, regional or national scale is required to allow policymakers and building energy experts to formulate energy policies for sustainable urban planning [30]. In addition, energy estimation helps evaluate the scope for energy retrofitting and reduce associated carbon emissions. Hence, archetypes are developed to simplify the analysis to represent a more extensive building stock. Typically, archetype characterisation at various spatial scales only uses national average U-values, which produce invalid energy results at district, city and regional scales [14,31,32]. Instead of directly using average values to fill the gaps in data, researchers use data-driven approaches to improve the quality of the data to develop a practical methodology for archetype development [33–35]. Machine learning techniques such as segmentation and clustering can be used on large datasets, including EPC published by the Sustainable Energy Authority of Ireland (SEAI) and

TABULA [36] to identify archetypes and required input parameters. A study by Sousa et al. used a data-tree structuring approach to represent archetypes with complex characteristics [37]. The number of archetypes developed depends upon the type of analysis and attributes considered during the segmentation of archetypes [38]. Ali et al. developed archetypes at multiple scales to predict the energy demand at various spatial scales by segmenting archetypes based on construction year and type of dwelling [14]. However, access to occupancy-related data is limited and therefore, modellers thus make assumptions for occupancy schedules based on prior knowledge, experience and available literature [19,39].

Using standardised occupancy schedules such as ASHRAE 90.1 [22] can be substantially different from the actual schedule patterns [40,41]. Du et al. 2020 [42] reviewed in-situ monitoring of occupant behaviour in residential buildings. The paper discusses the impact of occupant behaviour on building energy consumption and sustainable development and the effectiveness of in-situ monitoring using smart sensors in collecting unbiased data on occupant behaviour in the natural environment. Osman and Ouf 2021 [43] provide a comprehensive review of the use of time use surveys (TUS) in modelling occupant presence and behaviour for building energy research. Fewer studies focused on integrating realistic occupancy behaviour into archetypes. Buttitta et al. [13] developed and segmented occupancy integrated archetypes based on various occupancy schedules developed using TUS data, considering occupancy patterns as active or non-active. The profiles used a weekly average from Monday to Friday, separately for Saturday & Sunday throughout the year. Wolf et al. [44] used the Danish time-use survey data to develop stochastic occupancy profiles based on the Markov chain model. Even with certain assumptions, such as the location of occupants and type of house, the preliminary results agree with the measurements. Mitra et al. [45] developed stochastic occupancy profiles using American Time Use Survey data and reported that 41% variations in energy demand could occur in specific cases while using typical occupancy schedules. Sood et al. [46] determined the impact of the number of occupants on an apartment archetype. They concluded that a variation of 5%–8% can be observed primarily due to availability patterns determined based on various realistic activities performed by occupants daily. Aerts et al. [47] used hierarchical clustering to develop occupancy schedules based on 2005 Belgian time-use survey data. The occupancy patterns are divided into three states: (1) Awake and at home, (2) sleeping and (3) absent. These studies have found that variations in energy demand can occur in specific cases, with the percentage of variation varying from study to study. Diao et al. [22] used a bottom-up approach to modelling energy consumption in residential buildings based on occupant behaviour pattern clustering and stochastic simulation. The paper mentions that an unsupervised clustering algorithm was used to recognise and simulate occupant behaviour patterns but does not consider the impact of various parameters affecting their behaviour. Yao et al. [48] proposed a methodology framework for modelling occupants' stochastic behaviour on air conditioning in residential buildings. The framework combines measured data, statistical analysis, and logistic regression to derive stochastic behaviour models that can be incorporated into building simulation engines. However, the proposed method may not be accurate in dynamic and diverse communities where occupants have different lifestyles and behaviour patterns. Various large-scale empirical surveys were conducted in China to understand occupant behaviour in relation to residential energy use [49,50]. The study identifies typical patterns and characteristics of occupant behaviour, as well as factors that influence energy-related behaviour in residential buildings. Although these profiles were based on actual survey data, profiles do not capture the effect of seasonal & monthly variations, dwelling types and the number of occupants. Cuerda et al. [51] developed the occupancy availability patterns and investigated their influence on energy demand. They recorded a variation of 15% in heating and cooling energy consumption when realistic profiles were compared with the standard profiles. Panchabikesan et al. [52] developed a data-driven framework to extract the occupancy schedules when data is available and identify various input parameters that could represent the occupancy in dwellings. The framework was applied to the eight apartments in Lyon, France, where the data were collected using sensors. The results showed that to find the availability state of the occupant in a dwelling, the logistic model developed using plug load and CO_2 concentration shows better results in confusion matrix and misclassification error. O'Neil and Niu [53] showed that occupancy behaviour-related inputs could lead to about 4% variations in HVAC peak and annual demand. Gill et al. observed that occupant behaviour accounts for 51% and 37% of the variance in heat demand and electricity demand, respectively, for the same type of dwelling [54]. An et al. [55] performed a study in China to determine peak cooling load in residential buildings using an oversimplified occupancy behaviour model. They concluded that oversimplification could result in the design of an inefficient cooling system. In residential buildings, the existing occupancy models are based on time-inhomogeneous Markov Chain Monte Carlo techniques using sensor data [47,56,57]. The models usually capture the occupancy behaviour for an individual building instead of at various spatial scales. For instance, Gilani et al. [58] developed a stochastic model to evaluate the effect of occupancy on lighting energy consumption at various spatial scales. The results conclude that deterministic models at larger scales can perform reasonably.

Energy consumption patterns for dwellings are highly stochastic and vary considerably between different customers, primarily due to occupancy behaviour and their availability in a dwelling [59–61]. Various studies were carried out to develop different occupancy models using the Markov chain and they majorly focus on commercial buildings such as office spaces [62–65]. To generate load profiles, Yao et al. [66] categorised energy consumption determinants into two categories: physical and behavioural. Widen et al. [67] used time-use data to interpret occupants' usage patterns and appliance ratings to produce energy consumption profiles. The load profiles generated using meter data can be hourly/weekly/monthly, or 15 min, depending on the device's capability. Causone et al. [68] developed occupant profiles for a year using the electric load profiles to represent internal heat gains in the energy simulations. Razavi et al. [69] used the smart-meter data to develop a dynamic genetic programming-based process to detect occupancy in residential buildings. The model can detect the availability of occupants with the area under the receiver operating characteristic (AUROC) of 0.98. Sensors and smart meters help capture the load variations in a building and based on these variations, occupancy schedules can be developed using various predictive models [40,70–72]. These deterministic models are easy to implement and can be represented as fractions with values in the range [0, 1]. Chong et al. [73] used spatial occupancy data and applied a Bayesian calibration approach to reduce occupancy prediction errors from 37% to 24%. The IEA-EBC Annex 66 project

focused on setting up a platform to standardise a simulation methodology to incorporate occupancy and occupant behaviour into building energy modelling [74]. Several publications were produced under Annex 66 [75,76]. Recently, various other technologies such as Wi-Fi signals, thermostat data [77], location-based services data [78], mobile global positioning system data are used to detect the occupancy and their behavioural patterns.

Most studies have focused on developing occupancy models based on data collected from sensors and energy meters to track the presence and absence of occupants and represent the load profiles. However, load profiles alone do not provide information on the cause of energy consumption linked to occupants' activities or their location. Furthermore, previous studies have not adequately explained the behavioural changes among occupants and the variations in their daily activities due to the type of dwelling, the number of occupants residing there, and seasonal variability. Thus, it is essential to develop realistic profiles of household occupants to understand how deeper knowledge of occupancy affects energy consumption. Although the impact of occupancy on building energy consumption varies significantly at various spatial scales, most studies focus on individual rooms or buildings. While studying individual buildings has its advantages, it is associated with fewer constraints and easier monitoring of a single individual instead of multiple occupants at different spatial scales. Recent research suggests that appropriate occupancy models should be studied at various spatial and temporal resolutions [55]. Previous studies have linked occupancy to building energy modelling, but there is a need for more information on evaluating the influence of realistic occupancy behaviour on the energy consumption of archetypes at various spatial and temporal scales. Typically, building simulations use conventional occupancy schedules to simplify the process. However, this approach can lead to significant variations in output energy consumption, which can impact policy recommendations for retrofitting or other purposes. There is a significant difference between predicted and actual energy consumption, with a significant portion of this deviation due to occupancy and their behaviour. As occupants significantly influence energy consumption in buildings, energy usage is highly dependent on the number of occupants, their availability, and the type of dwelling they live in. Therefore, it is crucial to consider the realistic presence of occupants within archetypes. Given that energy policy implementations and decisions require detailed information to address future residential planning interventions, archetype models must be developed for more granular temporal and spatial resolution. Therefore, the classification of archetypes should consider the number of occupants within archetypes to account for the variations in energy estimation.

This paper addresses the above-mentioned research gap by proposing a novel framework for developing realistic occupancy schedules using TUS data in order to evaluate the impact of occupancy on estimating energy consumption of archetypes at various spatial and temporal scales. The study develops an easy-to-implement methodology to extract realistic occupancy profiles considering the impact of the type of dwelling, the number of occupants and seasonal changes to be representative of a large number of households. The study is further extended to develop zone-wise occupancy profiles based on activities mentioned in the survey. The profiles developed in this study are then used for energy simulations to determine the impact of realistic occupancy schedules on the energy consumption of archetypes.

3. Methodology

The methodology is divided into two sections: (1) Development of day-wise occupancy profiles and (2) Multi-scale Energy modelling (Fig. 1). The devised methodology is generic and can be used for similar studies that use Time Use Survey (TUS) data. Section 3.1 focuses on developing year-round occupancy schedules based on various activities performed by occupants that lead to comprehending their actual presence/absence in the dwelling. Section 3.2 adds these schedules into multi-scale archetypes to evaluate the impact of occupancy on estimated energy consumption at various spatial and temporal scales. In this study, the term "multi-scale" refers to the fact that archetypes were developed for different geographical scales based on the multi-scale U-values of various building envelope values, which were determined using energy performance certificates (EPCs) data [14].

3.1. Development of occupancy profiles

3.1.1. Collection of data

This step gathers all the necessary data to develop stochastic occupancy profiles and evaluate their impact on multi-scale building archetypes. The datasets such as EPC and TABULA include geometric and non-geometric data, weather data, survey data and census data. Survey data such as TUS are required to develop day-wise occupancy profiles as these data contain information related to various activities performed by residents in a dwelling. The TUS data consists of other information, including dwelling type, number of occupants, age of occupants, month and days, which is crucial to understand the cause of variations in occupancy activities. Geometric information consists of dwelling type, building envelope, type of heating system, energy use intensity and the number of floors, doors & windows. Non-geometric information contains building envelope & other building element U-values, properties of HVAC used in the building and year of construction. The geometric and non-geometric information can be collected from EPC [32]. Although, in this study, we took the required geometric data from Ali et al. [14], who developed the archetypes using EPC so that a constructive comparison can be provided in terms of energy estimation.

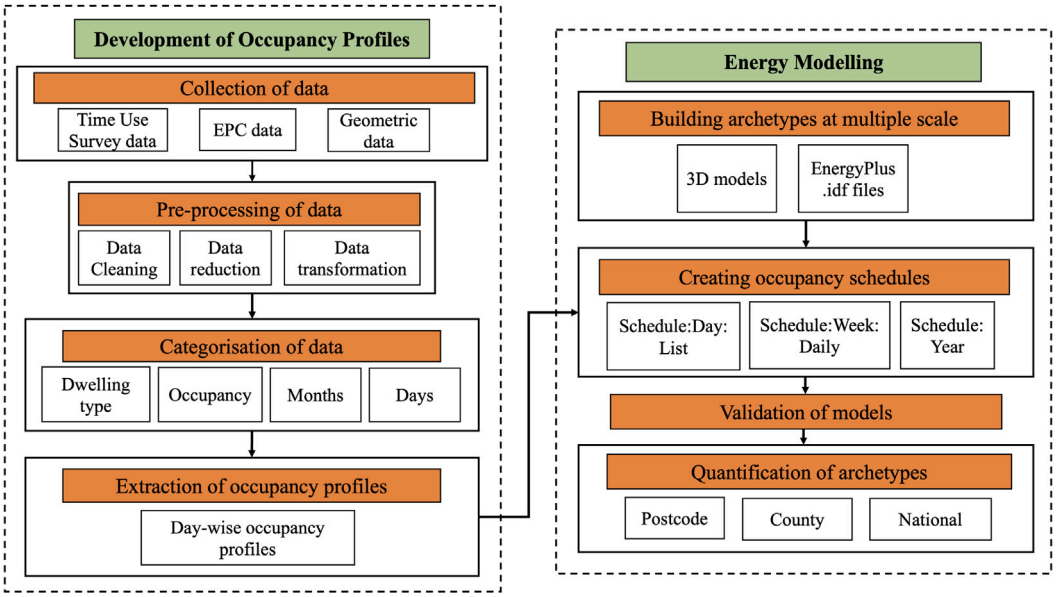


Fig. 1. Methodology to develop day-wise occupancy profiles and model them into archetypes to evaluate the impact of occupancy on multi-scale archetypes.

Table 1
Determining the location of activity based on the type of activity and typical location the activity can be performed.

Outdoor activity (OA)	Bedroom	Bathroom	Living Room	Kitchen
Office work				
Office break				
Admin work				
Travel				Cooking
Shopping	Sleeping		Reading	Dishwashing
Swimming	Cleaning	Toilet	Cleaning	Laundry
Running	Cloth	Cleaning	TV Watching	Food
Biking	management	Shower	Homework	management
Movie	Reading		Eating	Baking
Concert	Ironing			
Gym				
Gambling				
Fitness				
School				

3.1.2. Pre-processing of data

Data pre-processing consists of data cleaning, data reduction and data transformation. This study relies on the UK TUS data to develop occupancy schedules which are publicly available [79]. The data were recorded in a numeric format; every number represented the activity performed by the occupants. The occupancy data are usually collected through surveys prone to data irregularities such as incomplete and missing data. It is crucial to remove these inconsistencies in order to make the data usable. TUS data contains some entries that do not have all the information related to the occupancy activities. Therefore, rows with incomplete or non-applicable entries were removed from the dataset for data consistency throughout the analysis. Data reduction removes the irrelevant attributes from the dataset, such as year (IYear), day number (daynum), person number (pnum) and serial number. There were a few assumptions made before converting the TUS data into profiles:

1. The location of each activity was taken into consideration based on its nature. Based on the kind of activity, its location was determined (Table 1).
2. Activities that are carried out outside the dwelling, such as going to the gym, travelling, running, participating in sports, and engaging in social activities, banking, admin work, office work, movie, concerts, shopping, swimming and office break are categorised under the umbrella term “Outdoor Activities” or “OA”.

Based on the activity performed by residents, 0 and 1 are assigned to capture their absence/presence in a dwelling. For example, all the “OA” were assigned 0 and activities including sleeping, eating and bathing, i.e. all the indoor activities, are assigned 1. Internal loads are also modelled based on the developed profiles. The maximum gains values (e.g. people/m2, i.e. occupancy density) are multiplied by the values in the schedule to obtain the actual value to use at each time step in the simulation. We applied a

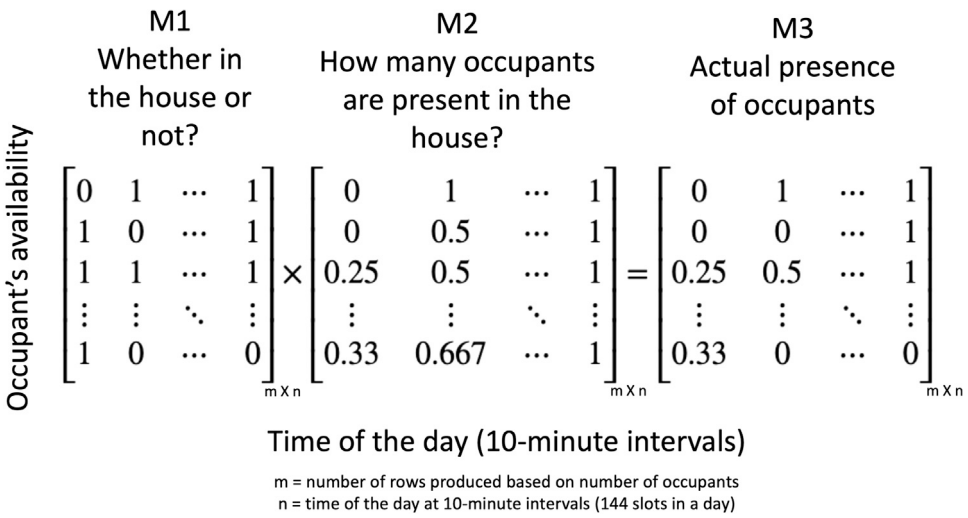


Fig. 2. Element-wise array multiplication to identify the actual presence of occupants in households with two or more occupants.

constant load between 8–23 W/m^2 depending upon the zone and equipment to account for the constant energy usage that occurs in a household by items such as a refrigerator and desktop computer.

Based on the assumptions, two arrays were created in order to extract the occupancy profiles. Element-wise array multiplication was employed (as shown in Fig. 2) to determine the presence of occupants in different zones within households with two or more occupants. The first array (M-1) represents the location of each occupant, i.e., whether they are in the bedroom, bathroom, kitchen or common room, with 1 indicating the presence and 0 indicating absence. The location of occupants is determined based on their activities and where they are being performed. For instance, in a single-occupant dwelling, if the occupant is present in the common room, the common room is assigned 1 and other zones are assigned 0. Similarly, for households with 2, 3 and 4 occupants, zones in which occupants are present are assigned 1 and others are assigned 0.

The second array (M-2) represents the total number of occupants present at a particular moment, irrespective of their location within the house. The dataset provides information on the number of occupants performing an activity with different people, such as alone, with a mother, with a spouse, with a father, and with a child, at 10 min intervals. Seven different datasets are provided based on the type of company. These datasets are added together to determine the total number of occupants present in the dwelling at 10 min intervals. To convert the data to a format that can be recognised by EnergyPlus, the added array is divided by the total number of occupants in the dwelling. For instance, if it is a 2-occupancy dwelling, only the dataset with 2 occupants is considered and divided by 2 after summing up all datasets with 2 occupants. The element-wise array multiplication provides the actual availability of occupants in different zones (M-3) within the range of [0, 1]. While M-1 is a variable matrix that changes with the zone, M-2 remains constant as it provides information on the total number of occupants present in the dwelling for a specific number of occupants.

3.1.3. Categorisation of data

This step categorises the data based on different attributes identified in the TUS data. The data classification initiated with the type of dwelling followed by the number of occupants, the month of the year and the day of the week (Algorithm 1). The process can be applied to any set of attributes depending upon the analysis. In this study, firstly, the data were grouped based on the accommodation type. We assumed that apartment-type dwellings such as top, mid % ground floor apartment, maisonette and basement dwellings shared similar characteristics as flats or maisonettes; therefore, the occupancy profiles developed for flats or maisonettes are also used for the various apartment types (Fig. 3). Similarly, occupancy profiles for house-type dwellings, such as semi-detached, detached, and terraced houses, used profiles developed for the houses or bungalows (Fig. 3). The total number of occupants in a household is determined by adding the total number of adults and children. This enables aggregation of the data based on the number of occupants living in a household. This allows us to observe the variations in occupancy behaviour and their presence/absence schedules. To further capture the effect of monthly and daily variations on occupancy activities, data were sorted into different months followed by days of the week. The steps mentioned above for categorisation enable us to capture various attributes' effects on occupancy behaviour (Algorithm 1).

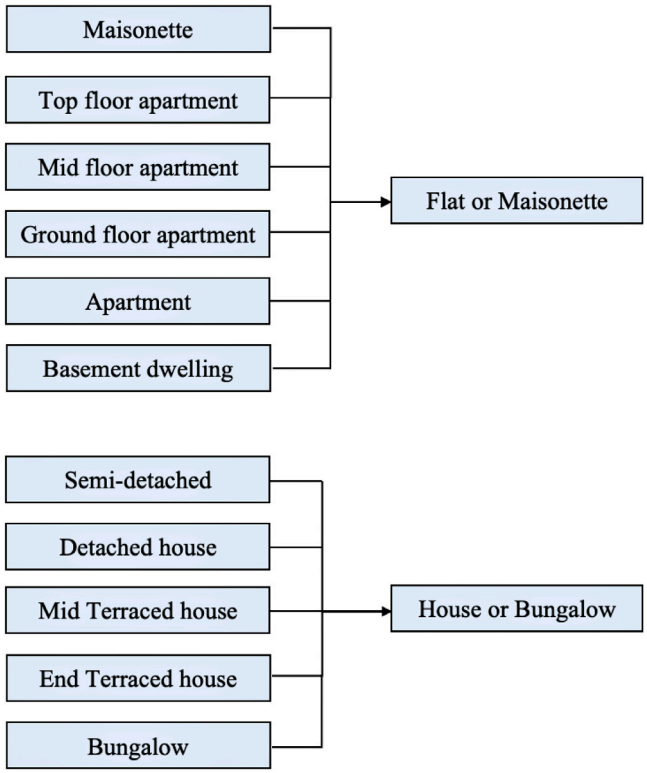


Fig. 3. Assumption to use developed occupancy profiles for various building archetypes based on the dwelling type mentioned in the TUS data.

Algorithm 1 Algorithm to make data clusters based on various attributes in order to extract desired occupancy profiles from TUS data

```
i = HouseOrBungalow, FlatOrMaisonette
for AccomType = i do
    for OccuNum = 1, 2, 3, 4 do
        for IMonth = 1, 2, 3 ... 12 do
            for DiaryDayAct = 1, 2, 3 ... 7 do
            end for
        end for
    end for
end for
DiaryDayAct.mode()
```

Once the data categorisation is completed, the next step is to extract the profiles. To represent time-use survey (TUS) data, categorical variables based on the number of states, such as active/inactive or active/inactive/absent, are commonly used. However, this limits the clustering algorithms that can be applied, with K-modes and Hierarchical clustering being the most suitable options [43]. These clustering algorithms have shown promising results in identifying activity patterns from TUS data, but explaining the various activities in a cluster remains a challenge as presence is usually categorised as active, non-active and absent. Therefore, the mode of the data is used in this study in order to consider the effect of type of dwelling, number of occupants, months of the year and day of the week on activity schedules. The mode helps in understanding the activities and assigning corresponding metabolic rates and energy system interaction so that realistic evaluation can be performed. The mode-based clustering is a better choice for clustering categorical data when simplicity, robustness, flexibility, and interpretability are important considerations [80]. We used the Statistics.mode() in R language to measure the central tendency in each time slot to extract the most frequently performed activities. Mode provides the activity that appears most often in each time slot, leading to a full-day activity profile. Initially, the data were divided into 10 min intervals; therefore, we have 144-time slots per day. The developed methodology also supports statistics.multimode(), i.e. bimodal or multimodal, to consider the second, third or further modes, if they exist.

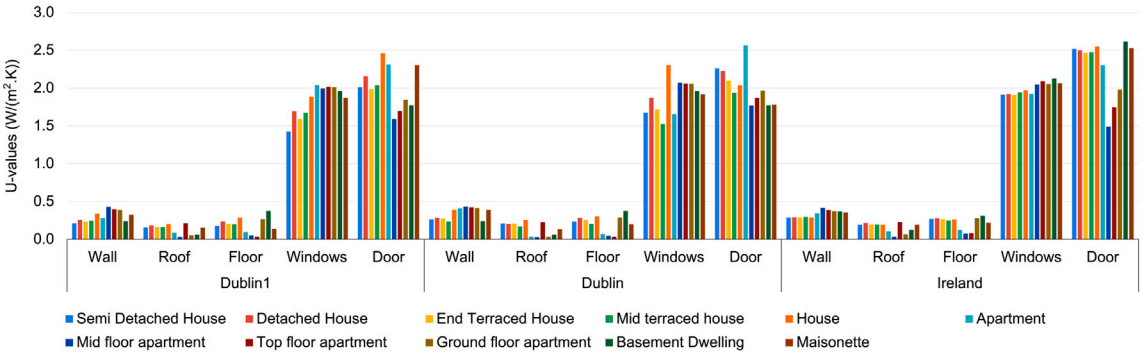


Fig. 4. Multi-scale U-values (W/m^2-K) for various building elements at postcode, county and national scales.

3.2. Multi-scale building energy modelling

This study uses archetypes to model energy consumption at different spatial scales, specifically focusing on electricity and heating usage, including space heating and domestic hot water. Some assumptions were made regarding domestic hot water, space heating, and lighting, but window opening and closing were not considered in this study. While these assumptions may not capture all possible variations in occupancy patterns and energy use behaviours, they are meant to represent typical residential energy use patterns and provide a simple and practical approach for modelling energy use in buildings. Assumptions are as follows:

1. For lighting, it was assumed that 1 h before sunset, the occupants switch ON lighting depending on the zone they are in. Occupants switch OFF lighting when they go to bed for sleeping. Therefore, lighting is ON from 1 h before sunset to 30 min after the first time sleeping appears in the activity diary at night. During the day, lights remain OFF. For lighting, it is common for occupants to turn on lights as daylight begins to fade and to turn them off when they go to sleep. By assuming that occupants turn on lights one hour before sunset and turn them off when they go to bed, we aimed to capture this common behaviour while keeping the lighting schedules simple.
2. For domestic hot water, the combination of kitchen and bathroom schedules is used since hot water is utilised in the kitchen and bathroom only. For domestic hot water, it is common for hot water to be used in the kitchen and bathroom, so we combined the schedules for these two zones to represent typical hot water usage patterns in a residential building.
3. For space heating, two hours in the morning (1 h during sleeping and 1 h after sleeping finished), 1 h in the evening when occupants are in the living room and 1 h when sleeping starts. For space heating, we assumed that occupants would typically turn on the heating system for a short period in the morning and evening when they are awake and active in the living room. We also assumed that the heating would be turned off during the day when occupants are likely to be out of the house and at night when they are sleeping. These assumptions were based on typical residential occupancy patterns and energy use behaviours.

3.2.1. Characterisation of archetypes

This step characterises the archetypes by using various data sources such as EPC's and TABULA to identify various information related to the building envelope. The U-values (W/m^2-K) for various building elements, including walls, floor, roof, doors and windows, were taken from [14] (Fig. 4) calculated for various spatial scales for different types of dwellings.

The above steps can be applied to any archetypes available in a particular region. For this study, in Ireland, 11 archetypes are developed based on the dwelling type and year of construction to represent the Irish residential building stock [14,38]. In this study, these archetypes are further improvised by classifying them based on the number of occupants. Fig. 5 illustrates the 3D models of the existing archetypes in Ireland. The first image in Fig. 5 represents all the apartment-type dwellings, namely, top, mid and ground floor apartments, basement dwellings and maisonette. The latter images represent house-type dwellings, including semi-detached, detached, bungalow or house and mid and & terraced hoses.

3.2.2. Adding occupancy and heat gains into archetypes

To model occupancy schedules for different building types, EnergyPlus objects such as Schedule:Day:List, Schedule:Week:Daily, and Schedule:Year were utilised. Weekly profiles were developed for each month, with a daily schedule assigned at 10 min intervals for each day of the week, resulting in seven profiles (Monday to Sunday) for each month. To create more detailed occupancy schedules, separate profiles were developed for each zone in the building, such as the kitchen, bedroom, bathroom, and common room, based on where occupants typically spend their time. If a dwelling had multiple bedrooms or toilets, the occupancy presence/absence value was divided by the number of that particular zone and assigned accordingly.

ASHRAE 55 provides metabolic rates (W/m^2) for typical tasks linked with occupancy in the same way occupancy schedules are linked with occupancy availability using detailed schedule EnergyPlus objects Schedule:Day:List. Zone-wise schedules were used along with corresponding metabolic rates to include the impact of various activities on energy consumption. For instance, various

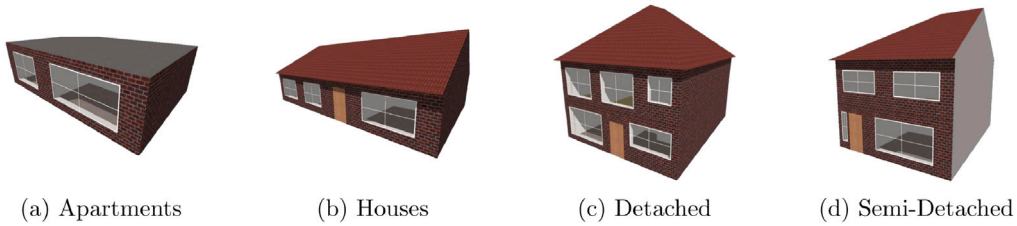


Fig. 5. 3D models of Irish building archetypes as a representative of Irish residential building stock for Energy Plus simulations.

activities are performed in a living room, so the average of all the activities was taken and used along with the living room schedule. Similar steps were followed for other zones and activities. For Bedroom (50 W/m^2), Living Room (92 W/m^2), Kitchen (115 W/m^2) and Bathroom (40 W/m^2) were used.

3.2.3. Quantification of archetypes

Archetypes help estimate the energy demand at various geographical scales without sufficient building data. The archetypes are modelled in Design Builder and simulated using EnergyPlus V9.6, the most commonly used software for BEPS. The occupancy profiles developed in this study are linked with the archetypes and parametric simulations were performed using the eplusr package for R scripts [81]. The energy analysis is carried out at various spatial scales, including postcode, county and national levels, using multi-scale residential building archetypes. The energy output at multiple scales allows us to quantify the variations due to occupants and their realistic behaviour.

3.2.4. Verification of results

The results of this study are validated using Dwelling Energy Assessment Procedure (DEAP) [82] Ireland's official method, to determine the BER of new and existing dwellings [38]. DEAP is a web-based platform used to calculate the annual delivered energy consumption, primary energy consumption (kWh/(m^2-yr)) and carbon dioxide emissions (kgCO_2/(m^2-yr)) for standardised occupancy [82]. The models used by Ali et al. [14] considered the conventional fixed, uniform and compact occupancy schedules provided by American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) [83] along with the same occupancy density at the building level. We used those models in this study and incorporated the developed occupancy schedules in order to evaluate the impact of occupancy on the energy consumption of archetypes. Since the input data, such as multi-scale U-values required for the energy simulations, remained the same except for the occupancy-related data, the variations that occur in the energy demand within a particular geographical scale are due to occupancy only.

4. Results

This section presents the findings of a study aimed at understanding the impact of occupancy on multi-scale residential building archetypes. The study involved developing day-wise occupancy profiles based on the various activities performed by residents and assessing the effect of occupancy on multi-scale Irish residential building archetypes. The data considered households with occupants ranging from 1 to 4 individuals, as these provided sufficient activity records. The developed profiles took into account the type of dwelling, number of residents, month of the year, and day of the week.

The study was conducted across cities in England, Scotland, Wales, and Northern Ireland, which are among the biggest and most densely populated regions in the UK. As the study did not distinguish between urban and rural areas, using these profiles to represent the whole country is a reasonable assumption. The survey was conducted in accordance with the guidelines of the Harmonised European Time Use Survey (HETUS), which is compatible with previous UK Time Use Studies and other European studies carried out since 2008. Hence, for this study on the Republic of Ireland, the UK TUS 2014–15 data was used.

4.1. Overview of the TUS data

The initial analysis of the UK TUS 2014–15 data provides insights into the wide range of activities performed by occupants in a household. The dataset consists of 16,550 diary days completed by respondents who were interviewed to complete two 24 h diary days. The activity diary records occupants' activities, such as sleeping, cooking, eating, bathing, and more, at 10 min intervals in a numeric code assigned to each activity. For instance, 110 represents sleep, 210 represents eating, 1110 represents working time in the main job, and 3130 represents dishwashing. A total of 11,860 sampled households resulted in 4238 household interviews with 10,208 eligible respondents, of whom 9388 answered the individual interview and/or completed 16,550 diary days. The diary also contains other information related to the type of housing, the number of adults and children in the house, the month of the year, and the day of the week the diary was filled. The data were recorded in such a way that 1 represents Sunday, 2 represents Monday, followed by the other days of the week in a similar pattern. The diary assigned an anonymised and unique serial number to every house which helped to identify the total number of respondents from the household. Fig. 6 displays the value count of the various activities recorded in the data by the respondents.

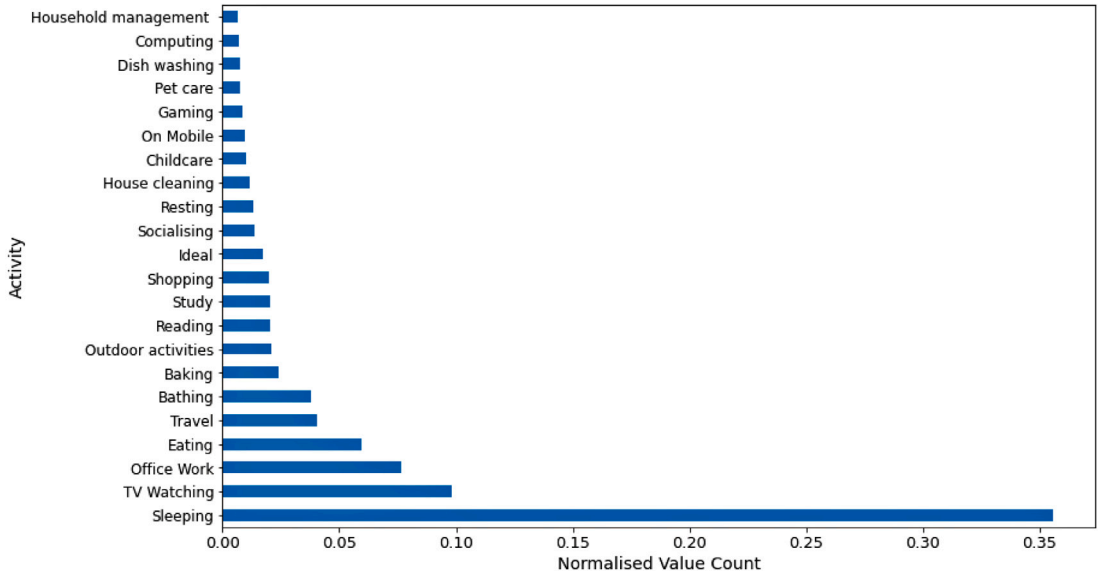


Fig. 6. Normalised value count of major activities mentioned in the TUS data to determine their frequency of occurrence.

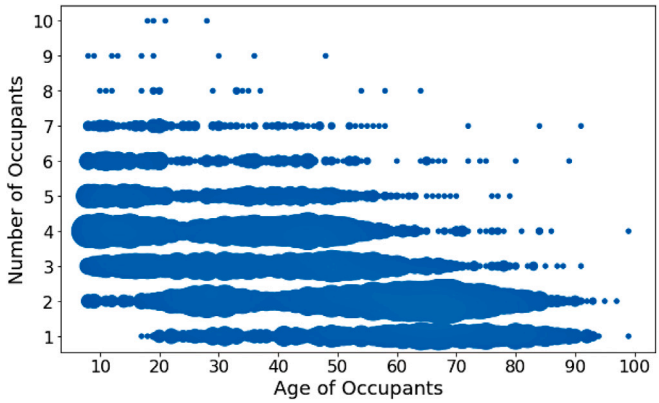


Fig. 7. Grouping of data based on the number of occupants to illustrate the age of the occupants who responded to the survey and filled the activity diary.

The data also contains information about where (wher_TimeSlot) and with whom (wit_TimeSlot) the activities were performed. As shown in Fig. 6, the data reveals that occupants engage in various activities on a daily basis, with sleeping having the highest value count, followed by TV watching and office work. The TUS data includes records of over 80 different activities performed by occupants, with only a few of them displayed in Fig. 6. The data also enables us to identify the location of each activity performed within a dwelling, allowing us to develop zone-wise profiles. It is worth mentioning that activities such as sleeping, office work, TV watching, eating, and travel are some of the most common activities performed by residents on a regular basis.

To gain a deeper understanding of the data, Fig. 7 provides information on the age of the respondents who participated in the survey and filled out the activity diary. The age range of the participants spans from as low as 10 to as high as 80 years, covering a diverse group of individuals with different work and living arrangements. It can be observed from Fig. 7 that for one- and two-occupant dwellings, the majority of the occupants are aged above 50 years. On the other hand, for 3- and 4-occupant dwellings, most occupants are within the age range of 30–50 years. The TUS data records occupants' activities effectively, enabling us to understand how people spend their time on a daily basis.

To begin with, the numeric codes in the data were replaced with their corresponding activities, as provided in the data dictionary. For some records that were missing, the average seasonal availability was used. For example, if values were missing for the month of May, the availability average for the spring months (March, April, and May) was taken for each time slot and used for further analysis. After grouping and filling in the missing data, activities were assigned values of 0 or 1 based on where they were performed, i.e., inside or outside the dwelling. Activities performed outside the dwelling, such as office work, running, shopping, etc., were assigned a value of 0, while those performed inside the dwelling, such as sleeping, eating, TV watching, etc., were assigned a value

of 1. However, to capture the occupancy presence in dwellings with a single occupant an element-wise matrix was then applied as described in Section 3.1.2 to determine the actual occupancy presence. The results are discussed in Section 4.2.

4.2. Categorisation and extraction of day-wise occupancy profiles

Sections 4.1 and 3.1.2 have provided insight into the TUS data used to develop day-wise occupancy schedules. This section demonstrates the occupancy profiles developed to consider daily and monthly variations for different types of dwellings with different occupancy. Apartment-type and house-type dwellings differ significantly in terms of structural and geometrical specifications. For this study, apartment-type dwellings represent top, mid, and ground-floor apartments, flats, and maisonettes, while house-type dwellings represent semi-detached, detached, and terraced houses, bungalows, and other types of houses. From Section 4.2 onwards, the results focus on one specific dwelling type, i.e., the semi-detached house. The plots provide information on weekly occupancy presence/absence for semi-detached houses over various months. Additionally, seasonal variations are presented due to the number of occupants in a dwelling. The profiles are demonstrated over a week to observe weekly variations.

These weekly profiles represent an entire month as the data mode is used to identify the most frequent daily activities. Therefore, one week's profile is used for other weeks in that month. Fig. 8 illustrates the weekly profiles of 1 occupant semi-detached dwelling for a particular month to represent the seasons in a year. These weekly profiles represent an entire month as the data mode is used to identify the most frequent daily activities. Therefore, one week's profile is used for other weeks in that month. January, April, July and October represent Winter, Spring, Summer and Autumn, respectively. This representation accounts for the seasonal variations and captures the daily occupancy activity discrepancies. For one occupancy dwelling, the resident's presence is represented by one and absence by 0. With various dissimilarities in the geometrical structure of dwellings, it is interesting to observe and quantify the effect of change in dwelling type on occupancy schedules. Fig. 8 demonstrates the variations in occupancy schedules for a one occupancy house for a particular month in fall under every season over the year. Compared with apartment-type dwellings (Fig. A.1), house-type dwellings (Fig. 8) depict significant variations in occupancy patterns. This variation due to dwelling type is usually missing from the existing research. For January, the occupant is absent approximately 11%, 13%, 20%, 5%, 26%, 11% and 14% on Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday, respectively. Likewise, an occupant is absent from the dwelling for approximately 18% of the week for April. Moreover, house-type dwelling profiles have more frequent inside and outside movement of residents, i.e. variation in presence/absence is more prominent compared to apartment-type dwellings (Fig. A.1). These stochastic profiles are more reasonable and justifiable than compact, uniform and fixed schedules for energy simulations. From Fig. 8 and Fig. A.1, it is evident that dwelling type is an important factor that influences the activities of the occupants. The space available in a dwelling largely defines the way occupants behave and impact their daily activities. Besides the dwelling type, seasonal change significantly influences their activities.

To comprehensively understand the effect of the number of occupants, Fig. 9 illustrates the occupancy schedules for the different numbers of occupants in a house-type dwelling for January. For one occupant dwelling, the occupancy presence/absence is denoted by discrete values 0 and 1. This means 1 denotes the occupant's presence in the dwelling and 0 relates to absence. However, for 2, 3 and 4 occupants' dwellings, apart from 0 and 1, occupancy presence/absence can take any values between 0 and 1 depending upon the total number of occupants residing in the dwelling. For instance, in two occupant dwellings, either both of them are present (1) or only one occupant is present (0.5), or none of them is present (0). Similarly, for three occupants at home, there is a possibility that all are present (1), one-third are present (0.33), two-thirds are present (0.67) and no one is present (0). For four occupants dwelling, the possibilities could be 0, 0.25, 0.5, 0.75 and 1. It can be noted from Fig. 9 that there is no day when the house remains vacant for 2, 3 and 4 occupants dwellings. There is always someone in the house performing some task that leads to energy consumption and should be considered to realistically estimate the energy demand. This provides an insight into the occupancy presence and their schedules instead of assuming fixed and uniform profiles throughout the year.

Fig. 10 depicts the yearly average occupancy availability schedules for various occupancy, i.e. variations in occupancy schedules due to the number of occupants in dwellings. The developed schedules are compared with the standard occupancy schedules for residential buildings provided by ASHRAE 90.1 and ASHRAE 62.1. The standard schedules are evidently different from the schedules developed in this study. When Fig. 10 is compared with Figs. 8 and 9, a significant variability in occupancy presence is observed. Hence, using average profiles throughout the year lead to significant variations in energy consumption. Therefore, it is recommended to use weekly or daily schedules in order to improve the accuracy of building performance analysis. Section 4.3 presents the energy consumption results based on developed stochastic occupancy availability schedules and compares them with the conventionally used schedules.

4.3. Archetype's energy modelling

This section presents an analysis of the variations in energy consumption of different building archetypes at various spatial scales due to realistic occupancy behaviour. The results are compared with the models used by Ali et al. [14] that used conventional occupancy schedules. The study focuses on providing energy consumption results at the archetype level only. All archetypes are developed at various spatial scales, including postcode, county, and national scales using Design Builder and are simulated in EnergyPlus. The occupancy-related data, including occupancy schedules along with occupancy density, were calculated based on different zones and the number of occupants.

This section provides a detailed analysis of energy use on an annual, monthly, and daily basis, particularly for the most common type of dwelling in Ireland, i.e. Semi-detached houses. The results obtained for a semi-detached house are validated using DEAP,

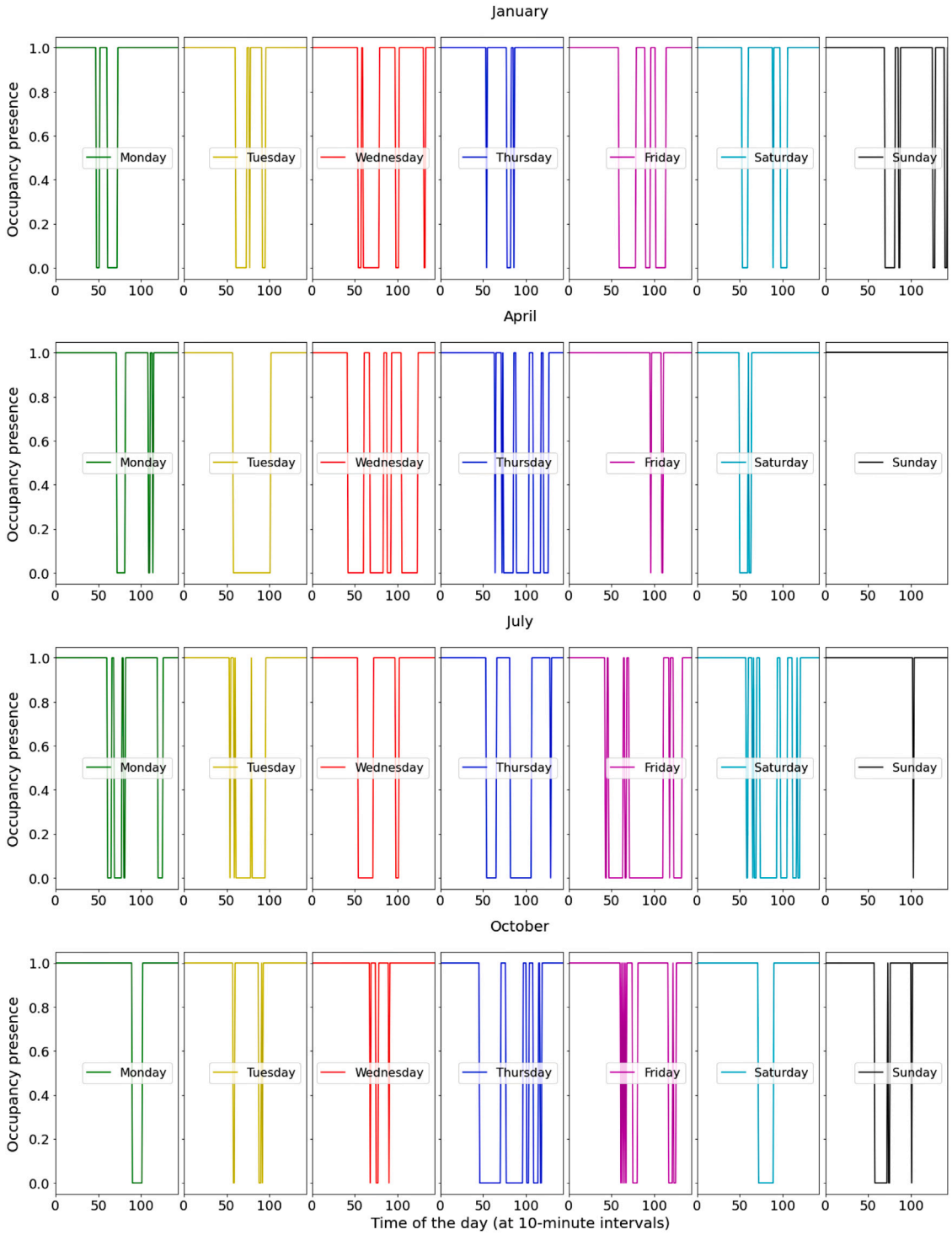


Fig. 8. Variations in occupant's availability (1 occupant) over a week in a house-type dwelling for months of January, April, July and October to represent Winter, Spring, Summer and Autumn, respectively.

which uses the national average occupancy as 2.88 ≈3. DEAP assumes the dwelling is used by standard or typical occupants and the heating system operating hours and temperature setpoints are fixed. Fig. 11 shows the monthly energy consumption (kWh) determined using DEAP and the results from the EnergyPlus simulations of the base case for the number of occupants as 3. The

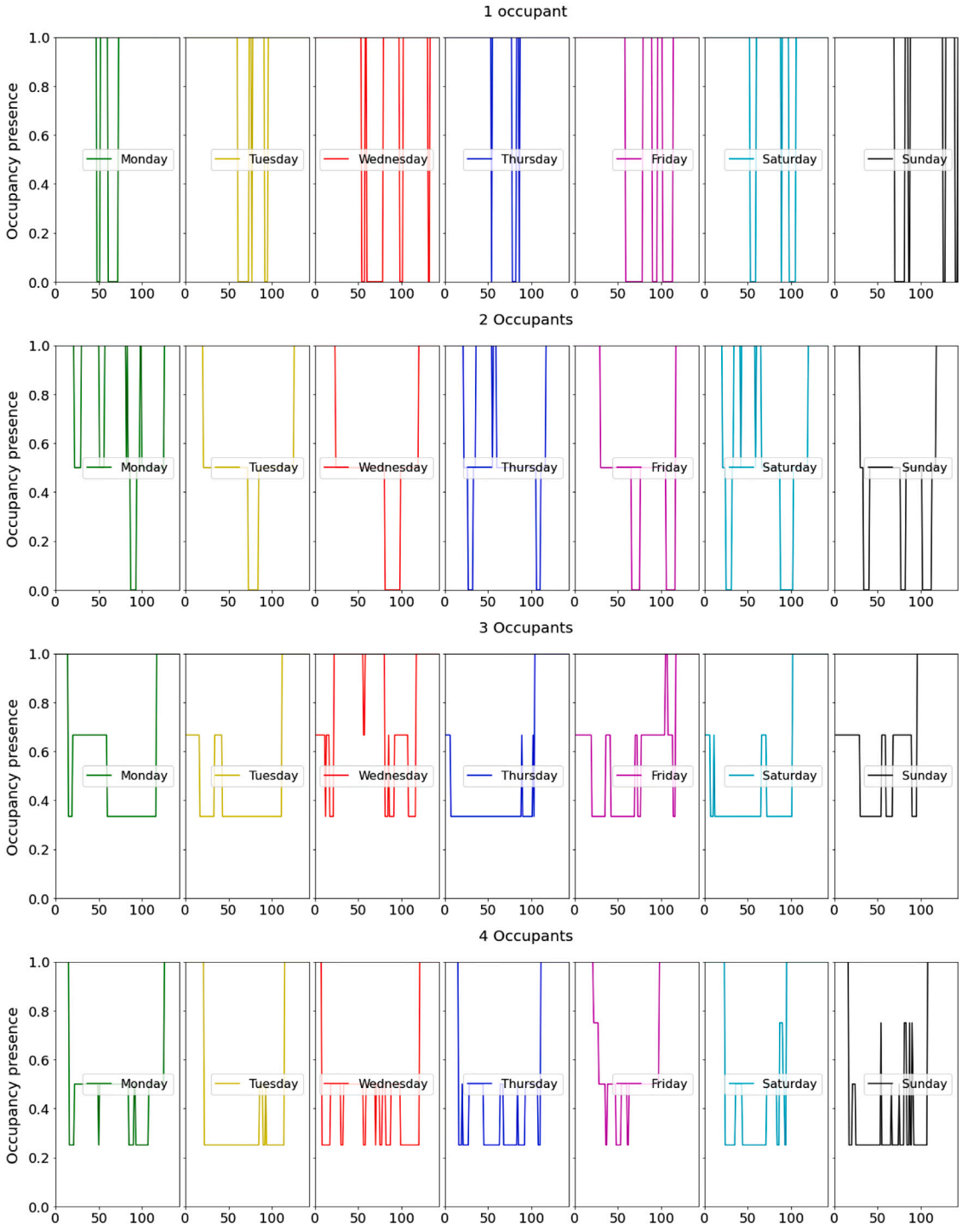


Fig. 9. Variations in occupancy absence/presence over a week due to the number of occupants in a house type dwelling for the month of January.

difference in the values appears due to the formulation of both software as DEAP is a static simulation software, whereas EnergyPlus is a dynamic simulation software. The inputs are the same for the DEAP and Energy Plus simulations, and both assume fixed and standard profiles for heating and setpoint temperature to match the criteria. Comparing the results produced from both software, an annual variation of 2.5% of the total energy consumption is reported for a semi-detached archetype. Similarly, for apartment archetypes, an annual variation of 6.3% of the total energy consumption is observed (Fig. 12). Moreover, for a semi-detached house,

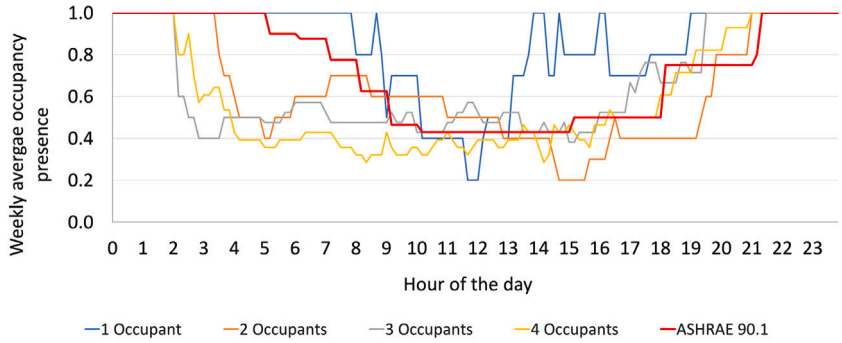


Fig. 10. Comparison of average availability schedules throughout the year for different occupancy in dwellings extracted from the TUS data with the standard schedules provided by ASHRAE.

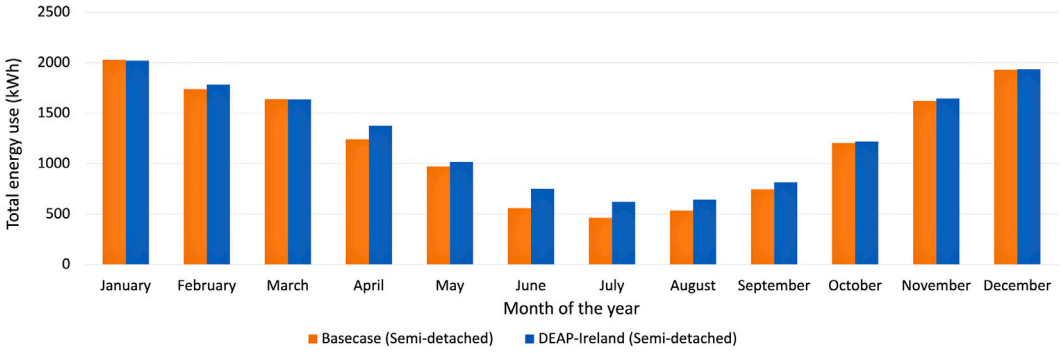


Fig. 11. Comparison of base case with Dwelling Energy Assessment Procedure (DEAP)-Ireland to validate the semi-detached archetype for effective analysis.

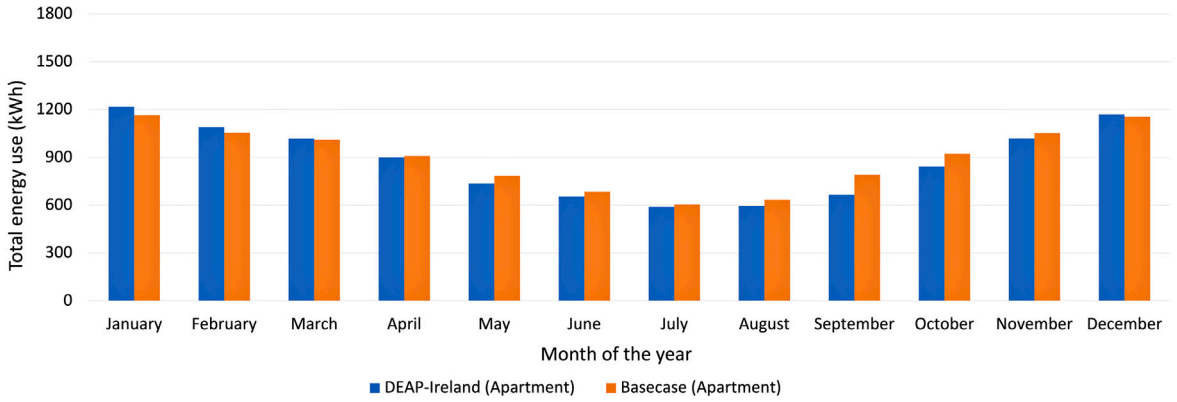


Fig. 12. Comparison of base case with Dwelling Energy Assessment Procedure (DEAP)-Ireland to validate the apartment archetype for effective analysis.

the average monthly difference between DEAP and EnergyPlus for a year is 73 kWh/yr (0.8%) of the total energy consumption and 37 kWh/yr (0.5%) for apartment archetypes. Therefore, the models can also be used for further analysis with varying occupancy.

4.4. Impact of occupancy on energy consumption of archetypes

The annual site energy consumption (kWh) for a semi-detached house at multiple scales, i.e. postcode, county and national level, is shown in Fig. 13. The results are compared with the base case from a recent study by Ali et al. [14], which used the conventional fixed and uniform schedules provided by the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) [83] for energy simulations. The present study shows an increasing trend in energy consumption when occupancy is taken into account and the number of occupants is increased from one to four. Energy consumption also increases

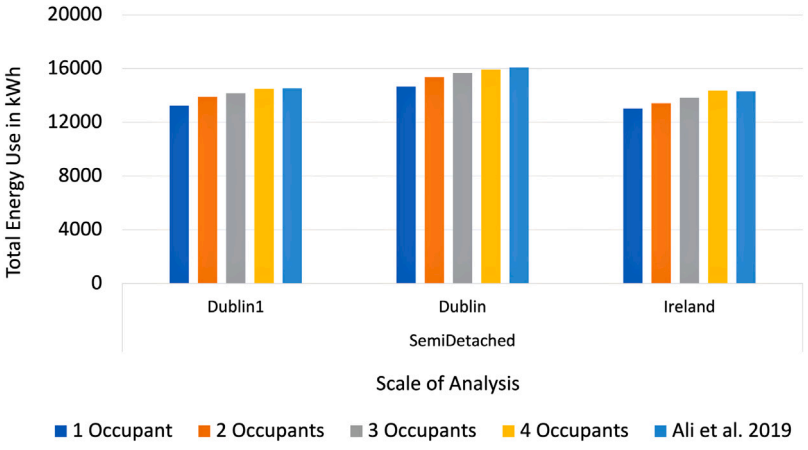


Fig. 13. Effect of occupancy on annual energy use (in kWh) in a Semi-detached house at multiple scales.

with the number of residents in a dwelling (Fig. 13). For total annual energy consumption values (kWh) for other archetypes at various spatial scales, refer to Figure A.2 in Appendix A.

At the postcode level i.e., Dublin1, the annual total energy consumption in a semi-detached dwelling with four occupants is 1238 kWh higher than that of a one-occupant dwelling. On average, an incremental difference of 413 kWh is observed in Dublin1 as the number of occupants increases from one to four, followed by 420 kWh at the Dublin level and 443 kWh at the national level. This corresponds to a variation of approximately 9.5% and 10.8% in annual energy use for Dublin and national scales, respectively. For further information on other archetypes, please refer to Table A.1 in Appendix A. When the base case archetype [14] is compared with one occupant semi-detached archetype at various spatial scales, an absolute difference of 1291 kWh, 1423 kWh and 1264 kWh at Dublin1, Dublin and Ireland, respectively, is observed. For a one-occupant semi-detached house, a variation in the range of 8.5%–10.3% (or 1264–1423 kWh) is noted when the scale of analysis changed from Dublin 1 to Ireland, i.e. from postcode to national scale, respectively. For more details on other archetypes, see A, Table A.2. Within the same geographical scale, variations in energy use are observed due to occupancy. However, when the scale of analysis is changed from Dublin1 to Dublin to Ireland, the variations in energy use are due to combined occupancy and geometrical characteristics.

Considering the high variability of occupancy and the number of occupants, it is reasonable to examine monthly and daily variations in energy consumption. Table 2 presents the mean, standard deviation, and absolute difference in monthly energy consumption due to the number of occupants. The results indicate that energy consumption follows the usual seasonal trend, with higher usage during the winter and lower usage during the summer. The data shows that the highest absolute difference due to occupancy is observed in January, with a difference of approximately 118 kWh when comparing a one-occupant dwelling to a four-occupant dwelling. This translates to an annual difference of 839 kWh. Overall, a 10% relative difference in combined heating and electricity consumption for a semi-detached house is observed due to the availability of occupants. Comparing the results to the base case archetype, which used conventional profiles, approximately 25% and 32% variations were observed in heating and electricity consumption, respectively (as shown in Table 3). Table 3 shows the differences in energy consumption for a semi-detached dwelling with varied occupancy for each month of the year compared to the base case. The differences are calculated by subtracting the energy consumption of the base case archetype from that of occupant-based archetypes. The results show that the differences vary throughout the year, with higher values in the colder months and lower values in the warmer months. For example, in January, the difference is 577 kWh, which is the second highest in the year after December. In contrast, in July, the difference is much lower, at 170 kWh. The standard deviation and mean values also vary throughout the year, with higher values in the colder months and lower values in the warmer months. These results suggest that occupancy significantly impacts energy consumption, with larger differences in energy use observed in the winter months when heating demands are higher.

On the other hand, the daily variations in energy consumption, particularly in electricity and heating, are quite different (as shown in Fig. 14). Electricity consumption in dwellings with three or four occupants is significantly higher than that in dwellings with one or two occupants because for 3 and 4 occupant households, there is always someone at home based on developed schedules. Additionally, in January, when the sun sets around 16:30, there are more dark hours and more occupants in a dwelling can result in increased electricity usage. The data indicate that energy consumption, particularly electricity usage, varies significantly depending on the number of occupants in a building. On average, buildings with more occupants consume more electricity than those with fewer occupants. There are some days in January where electricity consumption in a three-person dwelling is higher than that in a four-person dwelling by approximately 11.6%. However, there are notable daily variations, and there are some days where buildings with fewer occupants consume more electricity than those with more occupants. This could be due to differences in occupant behaviour and activities. The data from [14] serves as a reference point for comparison, and it shows that electricity consumption in the residential building in Dublin 1 is generally higher than the base case due to the more realistic occupancy schedules used in the present study. With regard to daily heating consumption, significant variations are observed due to realistic

Table 2
Absolute difference, standard deviation and mean of monthly total energy use (kWh) at postcode level due to the number of occupants, where OD is Occupant Dwelling and Δ denotes the difference between quantities.

Month	Δ (2OD-1OD)	Δ (3OD-2OD)	Δ (4OD-3OD)	Standard deviation	Mean
January	37	88	−7	62	1774
February	57	51	15	56	1630
March	69	42	27	60	1434
April	50	62	10	57	1078
May	28	67	4	49	841
June	26	38	10	34	549
July	16	23	−1	19	494
August	22	34	17	33	535
September	35	53	9	46	691
October	46	56	10	52	1079
November	80	41	34	67	1527
December	83	56	28	74	1795

Table 3
Absolute difference in monthly total energy use (kWh) at postcode level when variation in energy use due to occupancy is compared with the base case, where OD is Occupant Dwelling and Δ denotes the difference between quantities.

Month	Δ ([14]-1OD)	Δ ([14]-2OD)	Δ ([14]-3OD)	Δ ([14]-4OD)
January	577	539	451	457
February	515	459	407	392
March	561	493	450	423
April	415	365	303	293
May	383	356	288	285
June	224	199	161	151
July	170	154	130	131
August	226	204	171	153
September	283	248	195	186
October	387	340	284	274
November	517	437	396	362
December	590	507	451	423

occupancy schedules. As we move towards more granularity, the daily variations in heating and electricity consumption are 50% and 12%, respectively, when compared with conventional schedules (Fig. 14). An important finding is that energy usage, both in electricity and heating, tends to increase with the number of occupants in a building. This is evidenced by the fact that the mean values of both electricity and heating usage increase from one-person occupancy levels to four-person occupancy levels. Another interesting observation is that energy usage appears to vary considerably from day to day, which could be due to various factors such as changes in weather conditions, occupancy levels, or variations in occupants' behaviour, such as their use of appliances or heating controls. Also, with more people in the house, heating energy consumption is supposed to decrease because of more heat gain. However, this only happens when occupants are present in one zone only. Although with the developed profiles and the assumption regarding space heating, the majority of the time, occupants are present in different zones and using heating, thereby increasing the heating energy consumption.

5. Discussions

Multi-scale building energy analysis is a complex and computationally exhausting task [29]. Researchers made various assumptions to simplify the analysis, including uniform and fixed occupancy schedules. Most of the studies use schedules provided by ASHRAE 90.1 [47]. The study demonstrates that using typically fixed, uniform and compact occupancy schedules as input for dynamic building energy simulations is not usually representative of realistic occupancy schedules and thus reduces the accuracy and potential of predicting correct results. This study used the available UK TUS data to develop realistic daily, weekly and yearly occupancy availability schedules based on activities performed by occupants and evaluate their impact on energy consumption at various spatial and temporal scales using multi-scale residential building archetypes.

The occupancy is highly stochastic and can result in significant discrepancies in energy output by using fixed, uniform and compact profiles throughout the year. The TUS data record activities performed by residents, which help develop realistic occupancy presence/absence profiles. The data provides insight in to cause of energy use linked to activities performed by residents daily. Since these profiles are based on transforming the recorded activity data into binary data, thus the cause of their presence/absence is known. It is evident from the profiles (Figs. 8 and 9 that no day is the same in terms of availability of residents in a dwelling, therefore, assuming fixed availability led to a significant prediction error in energy estimation. For a one-occupant dwelling, either

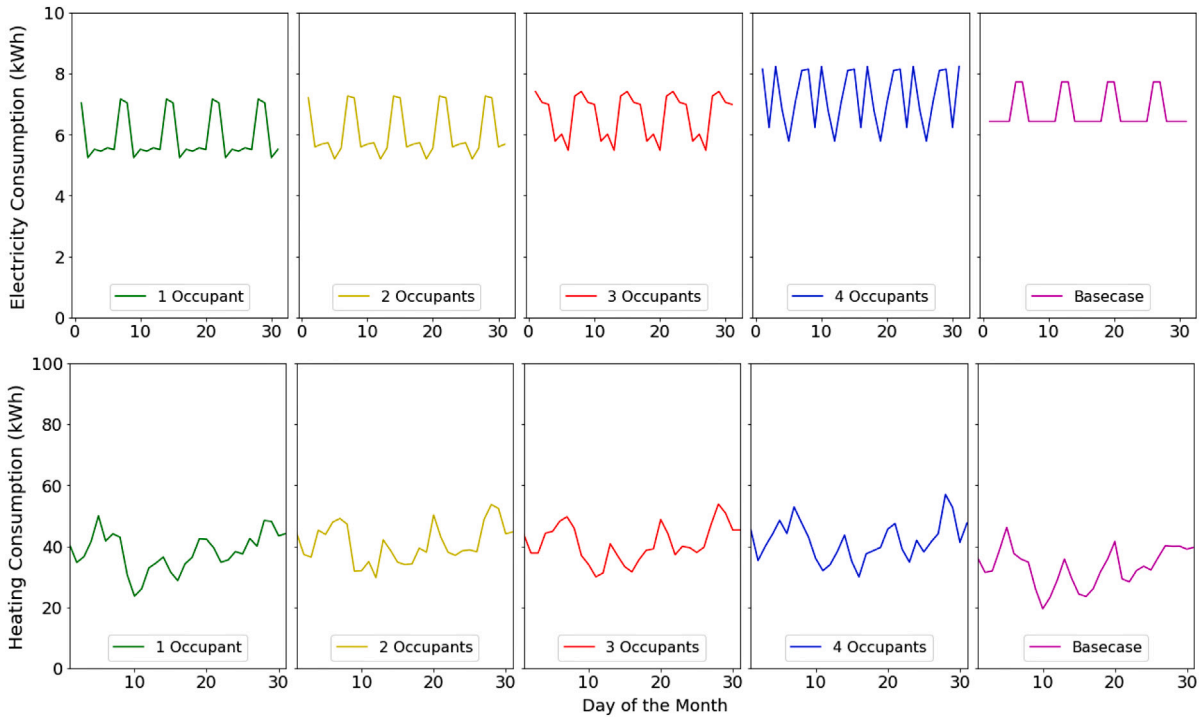


Fig. 14. Daily electricity and heating consumption patterns in a Semi-Detached House at postcode scale for different Occupancy (in kWh) for January.

the occupant is present or absent. On the other hand, for dwellings with 2, 3 and 4 occupants, there is no day in January that no one is absent from the home. Even with the change in season, the occupancy availability profile varies (Fig. 8). For a semi-detached house, approximately 12% and 18% of the time, the house remains vacant in January and April, respectively. Moreover, the type of dwelling also impacts the availability of residents. These profiles help understand the effect of various parameters, including the type of dwelling, number of occupants and day & month, on occupancy behaviour. With the improvement in results, these profiles could substitute for typically used fixed and uniform profiles.

Multi-scale building energy analysis is essential to identify the geographical areas with high energy consumption and allow policymakers and local authorities to amend policies and recommend suitable retrofit measures. The intended use is to analyse the daily, monthly and annual energy use (kWh) in residential buildings. These profiles significantly enhance the use of archetypes for the multi-scale building analysis as they help capture the diversity in energy use due to occupancy and their daily behaviour in residential buildings. The impact on daily, monthly and annual heating and electricity demand was determined, and we observed that occupancy significantly impacts the energy usage pattern. With the increase in occupancy, an increasing and convincing trend was perceived in heating and electricity consumption values. Approximately 8%–10% variation in annual total energy use (kWh) was observed when energy consumption in a one-occupant dwelling was compared with four-occupant dwellings. Moreover, the study compares the results with a previous study by Ali et al. [14] that used similar archetypes but with fixed and uniform profiles provided by ASHRAE. Compared with the base case, which used conventional shapes, approximately 25% and 32% variations were observed in heating and electricity consumption, respectively. As we are going into more granularity, the daily variations in heating and electricity consumption are 50% and 12%, respectively. It is worth mentioning that type of dwelling and kind of day & month can significantly influence the occupancy presence/absence in a dwelling. In addition, an elevated increment in electricity consumption in a semi-detached house with 4-occupants has been observed. This is because the frequency of staying at home in a 4-occupant semi-detached house is higher than 2 and 3 occupants dwelling, thus, higher electricity consumption (Figs. 9 and 14).

Therefore, it is necessary to develop models that reduce energy prediction errors so that archetypes facilitate appropriate policy recommendations based on which retrofit suggestions could be implemented. Timely adaption of energy efficiency measures helps in the sustainable development of the existing inefficient building stock. These occupancy-integrated archetypes improve the reliability of the results at various spatial and temporal scales and could help policymakers, local authorities and urban planners to modify, update and implement energy recommendations for ensuring direct retrofit investment and policies are based on actual data.

This work has certain limitations that need to be considered to enhance the results further. In the current study, only the first mode of the data is considered to extract the realistic presence/absence profiles. This does not consider other activities performed by occupants in the same period that could impact their availability in a dwelling. In addition, age is one factor that significantly impacts how residents behave in a dwelling. This study does not consider age as one of the parameters to categorise data. It is meaningful to consider age to observe the behavioural changes in activities performed by residents.

6. Conclusions and future work

This work aims to investigate the impact of realistic occupancy behaviour on multi-scale residential building archetypes, which can predict energy demand at different spatial and temporal scales. Typically, archetypes are developed without taking into account the realistic nature of occupants. This study made a few assumptions to determine the interaction of occupancy with lighting, space heating, and domestic hot water usage. The use of standard occupancy profiles throughout the year may lead to over or underestimation of energy demand. In this study, it was found that total energy consumption in a semi-detached dwelling is significantly different when varied occupancy and availability schedules are considered along with their stochastic profiles. Similar trends were observed for other dwelling types across various spatial and temporal scales. As we move towards more granularity, the impact of realistic schedules of occupants on energy usage increases, and the variations can be as large as 50% for heating and electricity consumption. Regarding total energy consumption, apartment-type dwellings were found to consume approximately four times less energy than house-type dwellings. This is due to the dwelling footprint and the distinction in activities performed in the respective dwelling types.

Based on the day-wise occupancy profiles, it is clear that no day is the same in terms of occupancy presence or absence in a dwelling. Moreover, the distribution of occupant availability varies across each season of the year. Another significant factor that impacts the activities performed by occupants is the type of dwelling. These profiles are representative of a large number of dwellings and indicate that, for two or more occupant dwellings, there is always someone present at home. The study demonstrates that using conventional and standard occupancy schedules as input for dynamic building energy simulations is not typically representative of realistic occupancy schedules, reducing the accuracy and potential for predicting correct results. Therefore, relying on conventional and standard occupancy availability schedules within archetypes leads to significant variations in energy predictions at various spatial scales. The integration of occupancy schedules into building archetypes enhances the reliability of results at various spatial and temporal scales. This approach could aid policymakers, local authorities, and urban planners in modifying, updating, and implementing energy recommendations based on actual data to ensure direct retrofit investment and policies.

The present work can be extended to develop a predictive occupancy model for energy simulations. The profiles can further be improved by considering other attributes, such as age and socio-economic status of occupants, which were not considered in this study. Uncertainty analysis can be added to the study to develop robust models. The activity data can also be mapped with the load profiles to understand occupancy activities and their effect on energy consumption. Moreover, IEQ is greatly influenced by the number of occupants and significantly affects energy consumption. Apart from the energy analysis, these realistic occupancy schedules allow us to determine the real overheating scenario and various indoor air quality parameters to appropriately evaluate the performance of various energy efficiency measures.

CRediT authorship contribution statement

Divyanshu Sood: Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft. **Ibrahim Alhindawi:** Writing – review & editing. **Usman Ali:** Acquisition of data. **James A. McGrath:** Writing – review & editing. **Miriam A. Byrne:** Writing – review & editing. **Donal Finn:** Writing – review & editing. **James O'Donnell:** Conception and design of study, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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